

rag_pipeline

May 29, 2026

1 AMR Policy Q&A — RAG Pipeline

A **Retrieval-Augmented Generation** system that answers natural-language questions about antimicrobial resistance policy, grounded in: - **WHO policy documents** (knowledge base, indexed in ChromaDB) - **Your XGBoost forecasting results** (injected as context per query) - **Phi-3 Mini** via Ollama (local LLM — no API keys needed)

User Question

ChromaDB Retriever WHO policy chunks (top-3 relevant)

Context Builder XGBoost forecast findings

Phi-3 Mini (Ollama) Grounded policy answer

1.1 1. Install Dependencies

```
[1]: # Run once - skip if already installed
import subprocess, sys

packages = ["chromadb", "sentence-transformers", "ollama", "pymupdf",
            ↪ "langchain-text-splitters"]
for pkg in packages:
    try:
        __import__(pkg.replace("-", "_"))
        print(f" {pkg} already installed")
    except ImportError:
        print(f"Installing {pkg}...")
        subprocess.check_call([sys.executable, "-m", "pip", "install", pkg,
            ↪ "-q"])
        print(f" {pkg} installed")

print("\n All dependencies ready!")
```

chromadb already installed

```
C:\Users\tanvi\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\site-packages\tqdm\auto.py:21:
TqdmWarning: IProgress not found. Please update jupyter and ipywidgets. See
https://ipywidgets.readthedocs.io/en/stable/user_install.html
```

```
from .autonotebook import tqdm as notebook_tqdm
```

```
sentence-transformers already installed
ollama already installed
pymupdf already installed
langchain-text-splitters already installed
```

All dependencies ready!

1.2 2. Import Libraries

```
[2]: import chromadb
from chromadb.utils import embedding_functions
from sentence_transformers import SentenceTransformer
import ollama
import pandas as pd
import numpy as np
import json, textwrap

print(" Libraries loaded!")
```

Libraries loaded!

1.3 3. Configuration

```
[3]: # Configuration
MODEL_NAME = "phi3:mini" # Ollama model - change freely
EMBEDDING_MODEL = "all-MiniLM-L6-v2" # Lightweight sentence embeddings
COLLECTION_NAME = "amr_policy_docs"
TOP_K = 3 # Chunks retrieved per query

# Verify Ollama is reachable
try:
    models = ollama.list()
    model_names = [m.model for m in models.models]
    if MODEL_NAME in model_names or any(MODEL_NAME in m for m in model_names):
        print(f" Ollama running | Model '{MODEL_NAME}' available")
    else:
        print(f" '{MODEL_NAME}' not found. Pull it with: ollama pull_
↳{MODEL_NAME}")
        print(f" Available models: {model_names}")
except Exception as e:
    print(f" Ollama not reachable: {e}")
    print(" Start Ollama desktop app, then re-run this cell.")
```

Ollama running | Model 'phi3:mini' available

1.4 4. WHO Policy Knowledge Base

Built-in starter knowledge base with key WHO AMR policy excerpts.

To add real PDFs: see the helper function at the bottom of this cell.

```
[4]: # Built-in WHO Policy Chunks
# Source: WHO Global Action Plan on AMR, WHO GLASS reports, WHO ESKAPE guidance
WHO_DOCUMENTS = [
    {
        "id": "who_gap_01",
        "title": "WHO Global Action Plan on AMR - Pillar 1: Awareness",
        "text": (
            "The WHO Global Action Plan on Antimicrobial Resistance (2015) identifies five strategic objectives. "
            "Pillar 1 focuses on improving awareness and understanding of antimicrobial resistance through "
            "effective communication, education, and training. Countries are urged to implement national "
            "communication strategies targeting healthcare workers, farmers, and the general public. "
            "Key interventions include restricting over-the-counter antibiotic sales without prescription, "
            "training community health workers on appropriate antibiotic use, and integrating AMR education "
            "into school curricula and professional training for healthcare providers."
        )
    },
    {
        "id": "who_gap_02",
        "title": "WHO Global Action Plan on AMR - Surveillance and Research",
        "text": (
            "Pillar 2 of the WHO Global Action Plan calls for strengthening knowledge and evidence base through "
            "surveillance and research. WHO's GLASS (Global Antimicrobial Resistance and Use Surveillance System) "
            "is the primary mechanism, enrolling countries to report standardized resistance data by pathogen, "
            "antibiotic, and specimen type. Priority pathogens under surveillance include E. coli, Klebsiella "
            "pneumoniae, Staphylococcus aureus, Streptococcus pneumoniae, Acinetobacter baumannii, and "
            "Neisseria gonorrhoeae. Countries with limited laboratory infrastructure are encouraged to start "
            "with sentinel surveillance sites before expanding nationally. "
        )
    }
]
```

```

    "Research priorities include new antibiotic development, rapid
↳diagnostics, and vaccine development."
  )
},
{
  "id": "who_escape_01",
  "title": "WHO ESKAPE Pathogens and Carbapenem Resistance",
  "text": (
    "WHO has designated carbapenem-resistant Enterobacteriaceae (CRE),
↳carbapenem-resistant Acinetobacter "
    "baumannii (CRAB), and carbapenem-resistant Pseudomonas aeruginosa
↳as CRITICAL priority pathogens "
    "requiring urgent research and new treatments. Carbapenems are
↳last-resort antibiotics; resistance "
    "renders infections nearly untreatable. WHO recommends strict
↳infection prevention and control (IPC) "
    "measures in hospitals, including contact precautions for CRE
↳carriers, environmental decontamination, "
    "and active surveillance cultures in high-risk units (ICUs,
↳oncology wards). "
    "For Klebsiella pneumoniae specifically, WHO recommends combination
↳therapy with colistin plus "
    "meropenem or tigecycline when carbapenem MICs are elevated. "
    "New beta-lactam/beta-lactamase inhibitor combinations
↳(ceftazidime-avibactam, meropenem-vaborbactam) "
    "are recommended where available. Low- and middle-income countries
↳face particular challenges "
    "in accessing these newer agents due to cost and availability
↳constraints."
  )
},
{
  "id": "who_stewardship_01",
  "title": "WHO AWaRe Classification and Antibiotic Stewardship",
  "text": (
    "WHO's AWaRe (Access, Watch, Reserve) classification categorizes
↳antibiotics to guide stewardship: "
    "ACCESS antibiotics are first/second-line treatments for common
↳infections (amoxicillin, ampicillin, "
    "gentamicin) and should be widely available. "
    "WATCH antibiotics have higher resistance potential and should only
↳be used for specific indications "
    "(fluoroquinolones, third-generation cephalosporins, carbapenems). "
    "RESERVE antibiotics are last-resort options (colistin, linezolid,
↳ceftazidime-avibactam) reserved "
    "for confirmed multi-drug-resistant infections. "
  )
}

```

```

        "WHO recommends that at least 60% of antibiotic consumption
        ↪nationally should come from Access group. "
        "Countries in South-East Asia and Africa show high Watch group
        ↪consumption, correlating with "
        "resistance emergence. Antibiotic stewardship programs (ASPs) in
        ↪hospitals reduce inappropriate "
        "prescribing by 20-30% when implemented with audit and feedback
        ↪mechanisms."
    )
},
{
    "id": "who_lmic_01",
    "title": "AMR in Low- and Middle-Income Countries - Surveillance
    ↪Investments",
    "text": (
        "Low- and middle-income countries (LMICs) bear disproportionate AMR
        ↪burden but face critical "
        "surveillance infrastructure gaps. WHO priorities for LMICs include:
        ↪ "
        "(1) establishing national reference laboratories with minimum AMR
        ↪testing capacity for priority pathogens; "
        "(2) training laboratory personnel in standardized susceptibility
        ↪testing (disk diffusion, MIC testing); "
        "(3) implementing electronic reporting systems linking hospitals to
        ↪national surveillance databases; "
        "(4) integrating AMR surveillance with existing disease
        ↪surveillance systems (e.g., IDSR in Africa). "
        "The African Region faces the highest mortality attributable to AMR
        ↪globally, with estimated "
        "230,000 deaths per year from drug-resistant infections. "
        "Investment priorities: point-of-care diagnostics, local antibiotic
        ↪production capacity, and "
        "community-based antibiotic stewardship in primary healthcare
        ↪settings. "
        "South-East Asia Region shows rapidly increasing carbapenem
        ↪resistance in hospital settings, "
        "driven by high antibiotic consumption and inadequate IPC in
        ↪resource-limited facilities."
    )
},
{
    "id": "who_resistance_trends_01",
    "title": "WHO GLASS Report - Regional Resistance Trends",
    "text": (
        "WHO GLASS data (2016-2022) reveals significant regional variation
        ↪in resistance patterns. "
    )
}

```

```

        "The European Region maintains the most comprehensive surveillance,
        ↳with the highest data quality, "
        "enabling reliable trend detection. Resistance rates for key
        ↳pathogens are generally lower in "
        "high-income European countries due to established stewardship
        ↳programs and regulatory frameworks. "
        "The Western Pacific Region shows heterogeneous patterns, with high
        ↳resistance in Southeast Asian "
        "nations contrasting with better-controlled rates in Australia, New
        ↳Zealand, and Japan. "
        "The Eastern Mediterranean Region faces dual challenges: high
        ↳consumption of Watch-group antibiotics "
        "combined with inadequate IPC infrastructure. "
        "In the African Region, limited GLASS participation means
        ↳surveillance gaps are significant, "
        "but available data indicates very high resistance rates
        ↳particularly for gram-negative pathogens. "
        "Countries with higher GDP per capita and universal health coverage
        ↳indices consistently show "
        "lower AMR rates, underlining the social determinants of
        ↳antimicrobial resistance."
    )
},
]

print(f" Knowledge base initialized with {len(WHO_DOCUMENTS)} WHO policy
↳documents")
for doc in WHO_DOCUMENTS:
    print(f" • [{doc['id']}] {doc['title'][:70]}...")

print(
    "\n" + "-"*50 + "\n"
    "To add real WHO PDFs, use this helper:\n"
    "    add_pdf_to_knowledge_base('path/to/who_doc.pdf')\n"
    "(defined at end of this cell)\n"
    + "-"*50
)

def add_pdf_to_knowledge_base(pdf_path, doc_id=None, chunk_size=500,
↳chunk_overlap=50):
    """Parse a PDF with PyMuPDF and chunk it with LangChain's
    RecursiveCharacterTextSplitter, then add chunks to WHO_DOCUMENTS.

```

Requirements (install once):

```
pip install pymupdf langchain-text-splitters
```

Args:

pdf_path : Path to the PDF file.

doc_id : Base ID for generated chunks (defaults to filename stem).

chunk_size : Target character size per chunk (default 500).

chunk_overlap: Overlap between consecutive chunks (default 50).

```
"""
# 1. Extract text with PyMuPDF (fitz)
try:
    import fitz # PyMuPDF
except ImportError:
    print("Install PyMuPDF first: pip install pymupdf")
    return

# 2. Semantic chunking with LangChain
try:
    from langchain_text_splitters import RecursiveCharacterTextSplitter
except ImportError:
    try:
        from langchain.text_splitter import RecursiveCharacterTextSplitter
    except ImportError:
        print("Install langchain-text-splitters first: pip install \
↳langchain-text-splitters")
        return

# Extract full text, preserving paragraph structure
doc = fitz.open(pdf_path)
pages_text = []
for page in doc:
    text = page.get_text("text") # plain text, layout-aware
    if text.strip():
        pages_text.append(text)
doc.close()
full_text = "\n\n".join(pages_text)

if not full_text.strip():
    print(f" No text extracted from {pdf_path}. Is it a scanned PDF?")
    return

# Split into semantically coherent chunks
splitter = RecursiveCharacterTextSplitter(
    chunk_size=chunk_size,
    chunk_overlap=chunk_overlap,
    separators=["\n\n", "\n", ". ", " ", ""], # prefer paragraph > \
↳sentence > word breaks
```

```

)
chunks = splitter.split_text(full_text)

# 3. Add chunks to WHO_DOCUMENTS
import os as _os
base_id = doc_id or _os.path.splitext(_os.path.basename(pdf_path))[0]
n = len(chunks)
for i, chunk in enumerate(chunks):
    WHO_DOCUMENTS.append({
        "id" : f"{base_id}_chunk_{i:03d}",
        "title": f"{base_id} (chunk {i+1}/{n})",
        "text" : chunk,
    })

print(f" Added {n} chunks from {pdf_path} "
      f"(avg {sum(len(c) for c in chunks)//max(n,1)} chars/chunk)")
print(" Re-run Section 5 to re-index the expanded knowledge base.")

```

Knowledge base initialized with 6 WHO policy documents

- [who_gap_01] WHO Global Action Plan on AMR - Pillar 1: Awareness...
- [who_gap_02] WHO Global Action Plan on AMR - Surveillance and Research...
- [who_eskape_01] WHO ESKAPE Pathogens and Carbapenem Resistance...
- [who_stewardship_01] WHO AWaRe Classification and Antibiotic Stewardship...
- [who_lmhc_01] AMR in Low- and Middle-Income Countries - Surveillance Investments...
- [who_resistance_trends_01] WHO GLASS Report - Regional Resistance Trends...

 To add real WHO PDFs, use this helper:

```

add_pdf_to_knowledge_base('path/to/who_doc.pdf')

```

(defined at end of this cell)

1.5 5. Embed Documents & Build ChromaDB Index

```

[5]: # GPU auto-detection with hardware probe
# torch.cuda.is_available() returns True even when the installed PyTorch build
# has no compiled kernel for the GPU architecture (e.g. RTX 5060 / sm_120).
# We run a tiny tensor op to confirm CUDA actually works before trusting it.
import torch as _torch

def _probe_cuda():
    """Return True only if CUDA is present AND a real op succeeds."""
    if not _torch.cuda.is_available():
        return False
    try:
        _t = _torch.tensor([1.0], device="cuda")
        _ = _t + _t # triggers kernel compilation / launch
    
```

```

        del _t
        return True
    except RuntimeError as _e:
        print(f"    CUDA detected but unusable ({_e.__class__.__name__}: {
↪str(_e)[:80]})")
        print(f"    Falling back to CPU - install a PyTorch build that supports
↪your GPU.")
        return False

_device = "cuda" if _probe_cuda() else "cpu"
if _device == "cuda":
    print(f"    GPU verified: {_torch.cuda.get_device_name(0)} - using CUDA for
↪embeddings")
else:
    print(f"    Using CPU for embeddings (stable, just slower)")

# Initialize embedding model
print(f"Loading embedding model (downloads ~80MB on first run)...")
embedder = SentenceTransformer(EMBEDDING_MODEL, device=_device)
print(f"    Embedding model loaded: {EMBEDDING_MODEL} (device: {_device})")

# Initialize ChromaDB (local, in-memory - persists for this session)
chroma_client = chromadb.Client()

# Drop existing collection if re-running
try:
    chroma_client.delete_collection(COLLECTION_NAME)
except:
    pass

collection = chroma_client.create_collection(
    name=COLLECTION_NAME,
    metadata={"hnsw:space": "cosine"}
)

# Embed and index all documents
print(f"\nIndexing {len(WHO_DOCUMENTS)} documents...")
texts      = [doc["text"]    for doc in WHO_DOCUMENTS]
ids         = [doc["id"]     for doc in WHO_DOCUMENTS]
metadatas   = [{"title": doc["title"]} for doc in WHO_DOCUMENTS]

embeddings = embedder.encode(texts, show_progress_bar=True).tolist()

collection.add(
    ids=ids,
    documents=texts,
    embeddings=embeddings,

```

```

        metadatas=metadatas
    )

    print(f"\n {collection.count()} chunks indexed into ChromaDB")
    print("Ready to retrieve WHO policy context!")

```

GPU verified: NVIDIA GeForce RTX 5060 Laptop GPU - using CUDA for embeddings
 Loading embedding model (downloads ~80MB on first run)..
 Embedding model loaded: all-MiniLM-L6-v2 (device: cuda)

Indexing 6 documents...

Batches: 100%| | 1/1 [00:00<00:00, 2.52it/s]

6 chunks indexed into ChromaDB
 Ready to retrieve WHO policy context!

1.6 6. Load Forecasting Results as Context

```

[6]: # Load model comparison results
import os as _os

_csv_path = "model_comparison_results.csv"

# Try to locate the CSV relative to the notebook or one directory up
for _candidate in [_csv_path, _os.path.join("files", _csv_path)]:
    if _os.path.exists(_candidate):
        _csv_path = _candidate
        break

try:
    results_df = pd.read_csv(_csv_path)
    print(f" {_os.path.basename(_csv_path)} loaded ({len(results_df)} rows)")
except FileNotFoundError:
    results_df = None
    print(f" model_comparison_results.csv not found - using fallback_
    defaults")

# Dynamically build FORECAST_CONTEXT from the CSV
def _fmt(val, decimals=2):
    """Format a numeric value or return a dash for NaN/missing."""
    try:
        return f"{float(val):.{decimals}f}"
    except (TypeError, ValueError):
        return "N/A"

```

```

if results_df is not None:
    # Identify best model by lowest Test MAE
    _test_mae_col = [c for c in results_df.columns if "test" in c.lower() and
    ↪ "mae" in c.lower()]
    _val_mae_col = [c for c in results_df.columns if "val" in c.lower() and
    ↪ "mae" in c.lower()]
    _test_r2_col = [c for c in results_df.columns if "test" in c.lower() and
    ↪ "r2" in c.lower()]

    _tcol = _test_mae_col[0] if _test_mae_col else None
    _vcol = _val_mae_col[0] if _val_mae_col else None
    _r2col = _test_r2_col[0] if _test_r2_col else None

    _valid = results_df.dropna(subset=[_tcol]) if _tcol else results_df
    _best = _valid.loc[_valid[_tcol].idxmin()] if _tcol else results_df.iloc[0]
    _naive = results_df[results_df["Model"].str.lower().str.contains("naive")].
    ↪ iloc[0] if results_df["Model"].str.lower().str.contains("naive").any() else
    ↪ None

    # Build model performance lines
    _perf_lines = []
    for _, row in results_df.iterrows():
        _mae_str = _fmt(row[_tcol]) if _tcol else "N/A"
        _r2_str = _fmt(row[_r2col], 3) if _r2col else "N/A"
        _perf_lines.append(f" - {row['Model']:<22} | MAE: {_mae_str}% | R²:
    ↪ {_r2_str}")

    _naive_mae = _fmt(_naive[_tcol]) if _naive is not None and _tcol else "41.
    ↪ 83"
    _best_mae = _fmt(_best[_tcol]) if _tcol else "N/A"
    _multiplier = f"{float(_naive_mae)/float(_best_mae):.0f}x" if _best_mae not
    ↪ in ("N/A",) else "N/A"

    FORECAST_CONTEXT = f"""
=== AMR FORECASTING RESULTS ({_best["Model"]}, Best Model) ===
Training data: WHO GLASS 2021-2023 | 44 countries | 5,909 observations

Model Performance (Test Set, Year 2023):
{chr(10).join(_perf_lines)}
- Naive baseline | MAE: {_naive_mae}% ({_multiplier} worse than
    ↪ {_best["Model"]})

Top Predictive Features:
1. Resistance_lag1 (50.5% importance) - prior year resistance is the
    ↪ strongest predictor
2. CountryTerritoryArea (9.0%) - country-level heterogeneity matters

```

- 3. AntibioticName (6.7%) - antibiotic-specific resistance patterns
- 4. PathogenName (6.0%) - pathogen identity is important
- 5. Antibiotic consumption DID (4.8%) - higher consumption correlates with ↴
↴higher resistance

MAE by WHO Region ({_best["Model"]} on 2023 test data):

- European Region : 4.2% (most accurate - best surveillance)
- Western Pacific Region : 7.3%
- African Region : 8.7% (sparse data = higher uncertainty)
- Eastern Mediterranean Region : 9.0%
- South-East Asia Region : 10.1% (highest resistance volatility)

Key Insight: Model accuracy strongly correlates with surveillance data quality. Regions with weaker GLASS participation have 2.4× higher forecast error.

"""

else:

Fallback - preserve original hardcoded context if CSV is unavailable

FORECAST_CONTEXT = """

=== AMR FORECASTING RESULTS (XGBoost, Best Model) ===

Training data: WHO GLASS 2021-2023 | 44 countries | 5,909 observations

Model Performance (Test Set, Year 2023):

- Best model: XGBoost | MAE: 7.07% | R^2 : 0.854
- Linear Regression | MAE: 8.23% | R^2 : 0.821
- LSTM (deep learning) | MAE: 7.16% | R^2 : ~0.85
- Naive baseline | MAE: 41.83% (55× worse than XGBoost)

Top Predictive Features:

- 1. Resistance_lag1 (50.5% importance) - prior year resistance is the ↴
↴strongest predictor
- 2. CountryTerritoryArea (9.0%) - country-level heterogeneity matters
- 3. AntibioticName (6.7%) - antibiotic-specific resistance patterns
- 4. PathogenName (6.0%) - pathogen identity is important
- 5. Antibiotic consumption DID (4.8%) - higher consumption correlates with ↴
↴higher resistance

MAE by WHO Region (XGBoost on 2023 test data):

- European Region : 4.2% (most accurate - best surveillance)
- Western Pacific Region : 7.3%
- African Region : 8.7% (sparse data = higher uncertainty)
- Eastern Mediterranean Region : 9.0%
- South-East Asia Region : 10.1% (highest resistance volatility)

Key Insight: Model accuracy strongly correlates with surveillance data quality. Regions with weaker GLASS participation have 2.4× higher forecast error.

"""

```
# Display results
if results_df is not None:
    print("\nModel Comparison Table:")
    print(results_df.to_string(index=False))

print("\n" + FORECAST_CONTEXT)
```

model_comparison_results.csv loaded (6 rows)

Model Comparison Table:

Model	Validation MAE	Test MAE	Validation RMSE	Test RMSE
Naive	41.827262	41.827262	48.149718	48.149718
Linear Regression	8.635278	8.234515	12.923747	11.962802
Ridge	8.639178	8.245513	12.898719	11.948087
XGBoost	7.092339	7.069361	11.623738	10.803452
LightGBM	7.226457	7.170185	11.810058	10.837808
LSTM	NaN	7.161261	NaN	10.097988

=== AMR FORECASTING RESULTS (XGBoost, Best Model) ===

Training data: WHO GLASS 2021-2023 | 44 countries | 5,909 observations

Model Performance (Test Set, Year 2023):

- Naive	MAE: 41.83%	R ² : nan
- Linear Regression	MAE: 8.23%	R ² : 0.821
- Ridge	MAE: 8.25%	R ² : 0.821
- XGBoost	MAE: 7.07%	R ² : 0.854
- LightGBM	MAE: 7.17%	R ² : 0.853
- LSTM	MAE: 7.16%	R ² : 0.872
- Naive baseline	MAE: 41.83% (6× worse than XGBoost)	

Top Predictive Features:

1. Resistance_lag1 (50.5% importance) - prior year resistance is the strongest predictor
2. CountryTerritoryArea (9.0%) - country-level heterogeneity matters
3. AntibioticName (6.7%) - antibiotic-specific resistance patterns
4. PathogenName (6.0%) - pathogen identity is important
5. Antibiotic consumption DID (4.8%) - higher consumption correlates with higher resistance

MAE by WHO Region (XGBoost on 2023 test data):

European Region	: 4.2%	(most accurate - best surveillance)
Western Pacific Region	: 7.3%	
African Region	: 8.7%	(sparse data = higher uncertainty)
Eastern Mediterranean Region	: 9.0%	
South-East Asia Region	: 10.1%	(highest resistance volatility)

Key Insight: Model accuracy strongly correlates with surveillance data quality. Regions with weaker GLASS participation have 2.4× higher forecast error.

1.7 7. RAG Query Function

```
[7]: def retrieve_context(question, top_k=TOP_K):
    """Retrieve top-k relevant WHO policy chunks from ChromaDB."""
    query_embedding = embedder.encode([question]).tolist()
    results = collection.query(
        query_embeddings=query_embedding,
        n_results=top_k,
        include=["documents", "metadatas", "distances"]
    )
    chunks = results["documents"][0]
    titles = [m["title"] for m in results["metadatas"][0]]
    distances = results["distances"][0]
    return chunks, titles, distances

def build_prompt(question, retrieved_chunks, retrieved_titles):
    """Build the full prompt: system context + WHO chunks + forecast data +
    question."""
    who_context = ""
    for i, (chunk, title) in enumerate(zip(retrieved_chunks, retrieved_titles)):
        who_context += f"\n[Source {i+1}]: {title}]\n{chunk}\n"

    prompt = f"""You are an expert in antimicrobial resistance (AMR) policy and
    public health.
    Answer the user's question using ONLY the WHO policy excerpts and forecasting
    data provided below.
    Be specific, cite which WHO guidance applies, and connect forecasted resistance
    trends to policy recommendations.
    If the answer cannot be found in the provided context, say so clearly.
    STRICT RULES:
    - Only cite sources as [Source 1], [Source 2], [Source 3] etc. - never invent
    URLs, journal names, or publication links.
    - Do NOT fabricate statistics, citations, or external references not present in
    the context.
    - If unsure, say "the provided context does not specify this.""""
```

```

=== WHO POLICY CONTEXT ===
{who_context}

=== FORECASTING DATA ===
{FORECAST_CONTEXT}

=== USER QUESTION ===
{question}

=== YOUR ANSWER ===
"""
    return prompt

def ask_amr(question, verbose=False):
    """Full RAG pipeline: retrieve → build context → query Phi-3 Mini → return
    ↪ answer."""
    print(f"\n{'='*65}")
    print(f" {question}")
    print(f"{'='*65}")

    # Retrieve
    chunks, titles, distances = retrieve_context(question)
    if verbose:
        print("\n Retrieved sources:")
        for t, d in zip(titles, distances):
            print(f"    [{d:.3f}] {t[:70]}...")

    # Build prompt
    prompt = build_prompt(question, chunks, titles)

    # Query Phi-3 Mini via Ollama
    try:
        response = ollama.chat(
            model=MODEL_NAME,
            messages=[{"role": "user", "content": prompt}]
        )
        answer = response["message"]["content"]
    except Exception as e:
        answer = f" Ollama error: {e}\n    Make sure Ollama is running and
        ↪ '{MODEL_NAME}' is available."

    # Print formatted answer
    wrapped = textwrap.fill(answer, width=80, subsequent_indent="    ")
    print(f"\n {wrapped}\n")

```

```

return answer

print(" RAG query function ready!")
print(f" Model: {MODEL_NAME}")
print(f" Retrieval: top-{TOP_K} chunks from {collection.count()} indexed_
↳documents")

```

```

RAG query function ready!
Model: phi3:mini
Retrieval: top-3 chunks from 6 indexed documents

```

1.8 8. Demo — 5 Sample Questions

```

[8]: # Run all 5 demo questions
DEMO_QUESTIONS = [
    "Which antibiotics should Southeast Asia prioritize preserving based on_
↳resistance forecasts?",
    "What does WHO recommend for treating carbapenem-resistant Klebsiella_
↳pneumoniae?",
    "Which WHO regions are forecasted to have the highest resistance_
↳uncertainty and why?",
    "How do resistance trends differ between high-income and low-income_
↳countries?",
    "What surveillance investments does WHO prioritize for low-resource_
↳settings?",
]

answers = {}
for q in DEMO_QUESTIONS:
    answers[q] = ask_amr(q)

```

```

=====
Which antibiotics should Southeast Asia prioritize preserving based on
resistance forecasts?
=====

```

The Southeast Asia Region should prioritize preserving antibiotics from the ACCESS group, according to the WHO AWaRe classification. Source 3 emphasizes the importance of ACCESS antibiotics as first/second-line treatments for common infections. These antibiotics, such as amoxicillin, ampicillin, and gentamicin, are crucial for addressing the most frequently occurring infections in the region. Additionally, Source 2 highlights the necessity of establishing national reference laboratories with minimum AMR testing capacity for priority pathogens in Southeast Asia, suggesting that the region should focus on antibiotics that are relevant for these pathogens. Moreover, Source 3 advises that at least 60% of antibiotic consumption nationally

should come from Access group. This underscores the need to reduce reliance on the WATCH group, particularly fluoroquinolones, third-generation cephalosporins, and carbapenems, as indicated by the high Watch group consumption in the region. Furthermore, Source 1 mentions the heterogeneous patterns in the Western Pacific Region and the high resistance in Southeast Asian nations, reinforcing the need for targeted stewardship programs in Southeast Asia. Source 3 also indicates that antibiotic stewardship programs (ASPs) in hospitals reduce inappropriate prescribing by 20-30% when implemented with audit and feedback mechanisms. Therefore, Southeast Asia should prioritize ASPs to preserve antibiotics and reduce misuse. In conclusion, Southeast Asia should prioritize preserving ACCESS group antibiotics and implementing antibiotic stewardship programs, particularly focusing on ASPs in hospitals, to address the resistance forecast and maintain the effectiveness of antibiotics in the region.

=====

What does WHO recommend for treating carbapenem-resistant *Klebsiella pneumoniae*?

=====

According to [Source 1: WHO ESKAPE Pathogens and Carbapenem Resistance], for carbapenem-resistant *Klebsiella pneumoniae*, WHO recommends using combination therapy with colistin plus meropenem or tigecycline when the carbapenem minimum inhibitory concentrations (MICs) are elevated. This guidance is particularly relevant for treating infections where carbapenems, the last-resort antibiotics, are ineffective due to resistance. [Source 1] also mentions that new beta-lactam/beta-lactamase inhibitor combinations, such as ceftazidime-avibactam and meropenem-vaborbactam, are recommended where they are available. However, their accessibility is limited in low- and middle-income countries due to cost and availability constraints. The provided context does not specify what WHO recommends for treating carbapenem-resistant *Klebsiella pneumoniae* outside of the mentioned context (no specific guidelines for this pathogen combination). The forecasting data suggests that country-level surveillance data quality directly impacts forecasting accuracy ([Source 3: WHO AWaRe Classification and Antibiotic Stewardship] and the top predictive features). This emphasizes the importance of robust surveillance and stewardship programs to inform treatment recommendations effectively and combat AMR. The AMR forecasting results demonstrate that the African and South-East Asia Regions have higher resistance levels, which correlates with higher forecast errors and underscores the urgency of improving surveillance and stewardship efforts to control AMR in these areas.

=====

Which WHO regions are forecasted to have the highest resistance uncertainty and why?

=====

According to the forecasting data provided by the XGBoost model on 2023 test data, the African Region is forecasted to have the highest resistance uncertainty with a Mean Absolute Error (MAE) of 8.7%. This high level of uncertainty can be attributed to the fact that the African Region has the lowest number of GLASS participations, as stated in the WHO policy context (Source 1). The limited surveillance data available for this region leads to a less accurate forecasting model, as the model's accuracy is strongly correlated with surveillance data quality (Key Insight). Additionally, the WHO policy context (Source 1) indicates that in regions with limited GLASS participation, there are significant surveillance gaps, further contributing to higher forecast error.

=====

How do resistance trends differ between high-income and low-income countries?

=====

High-income countries in the European Region exhibit lower antimicrobial resistance rates for key pathogens, as highlighted in [Source 1: WHO GLASS Report - Regional Resistance Trends]. This can be attributed to the establishment of stewardship programs and regulatory frameworks in these countries, which are not mentioned explicitly for low-income countries but can be implied to be less comprehensive or effective. The African Region, in contrast, shows very high resistance rates, particularly for gram-negative pathogens, as noted in the same source. The key insight here connects to [Source 2: WHO Global Action Plan on AMR - Pillar 1: Awareness], where it suggests the need for targeted national communication strategies and interventions in LMICs to improve antimicrobial stewardship, which aligns with the forecast that poorer surveillance infrastructure, as mentioned in [Source 3: AMR in Low- and Middle-Income Countries - Surveillance investments], contributes to these higher resistance rates. Furthermore, the AMR forecasting data indicates that regions with weaker GLASS participation, which are likely low-income countries, have 2.4× higher forecast error, directly linking lower data quality to higher uncertainty in resistance trends. This suggests that resistance trends are generally more unfavorable in low-income countries, potentially due to less effective stewardship and surveillance, as per [Source 3: AMR in Low- and Middle-Income Countries - Surveillance investments]. The data and policies underscore the need for enhanced surveillance and stewardship measures in low-income countries to manage AMR effectively.

=====

What surveillance investments does WHO prioritize for low-resource settings?

=====

According to [Source 2: AMR in Low- and Middle-Income Countries - Surveillance

Investments], WHO prioritizes the following surveillance investments for low-resource settings: 1. Establishing national reference laboratories with minimum AMR testing capacity for priority pathogens. 2. Training laboratory personnel in standardized susceptibility testing methods, such as disk diffusion and MIC testing. 3. Implementing electronic reporting systems that connect hospitals directly to national surveillance databases. 4. Integrating AMR surveillance with existing disease surveillance systems, for example, the Integrated Disease Surveillance and Response (IDSR) system in Africa. These investments aim to improve AMR surveillance infrastructure and capabilities in LMICs, addressing a critical gap that contributes to the disproportionate AMR burden they face.

1.9 9. Interactive Q&A

Type any AMR policy question. Type quit to exit.

```
[ ]: print("AMR Policy Q&A System - powered by Phi-3 Mini + WHO Documents + XGBoost_
↳Forecasts")
print("Type 'quit' to exit | Type 'verbose' before query for source details\n")

while True:
    try:
        user_input = input("Your question: ").strip()
    except (EOFError, KeyboardInterrupt):
        print("\nSession ended.")
        break

    if not user_input:
        continue
    if user_input.lower() == "quit":
        print("Session ended.")
        break

    verbose = False
    if user_input.lower().startswith("verbose "):
        verbose = True
        user_input = user_input[8:].strip()

    ask_amr(user_input, verbose=verbose)
```

AMR Policy Q&A System - powered by Phi-3 Mini + WHO Documents + XGBoost
Forecasts
Type 'quit' to exit | Type 'verbose' before query for source details